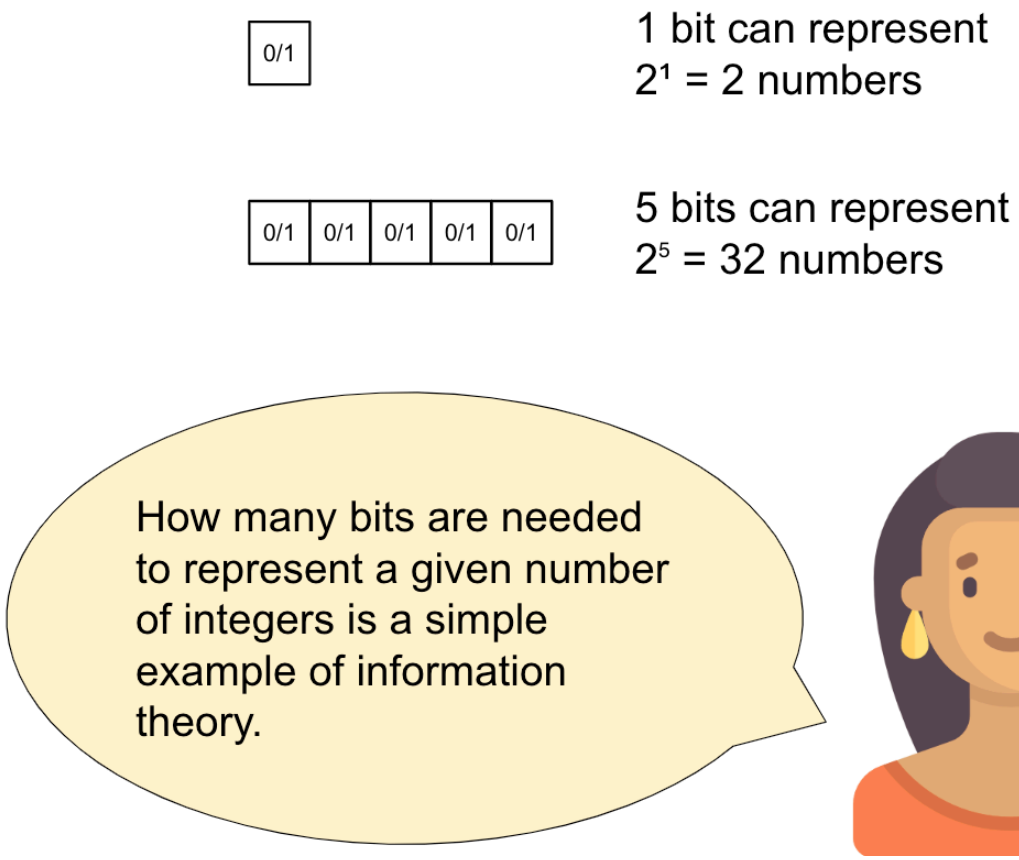


The information-theoretic concept of interest within the context of decision making under uncertainty is *entropy*. Entropy measures the level of uncertainty we have in an unknown quantity. If this unknown quantity is modeled as a random variable, entropy measures the variability in the possible values of the random variable.

> **NOTE** **This uncertainty measure, entropy, is similar to but not quite the same as what we've been calling uncertainty in the predictions made by a Gaussian process thus far, which is simply the standard deviation of the predictive distribution.**

In this subsection, we will be learning more about entropy as a concept and how it is computed for a simple Bernoulli distribution for binary events. We will see how entropy successfully quantifies uncertainty in an unknown quantity.

Let's go back to the first example of any Introduction to Probability course: coin flipping. Say you are about to flip a biased coin that will land on heads with some probability p in $(0, 1)$, and you'd like to reason about the event that the coin will indeed land on heads. Denote X to be the binary random variable that indicates whether the event happens (that is, $X = 1$ if the coin lands on heads and $X = 0$ otherwise). Then, we say that X follows a Bernoulli distribution with parameter p, and the probability that $X = 1$ is equal to p.

©Manning Publications Co. To comment go to liveBook
https://livebook.manning.com/#!/book/bayesian-optimization-in-action/discussion

Licensed to ashwin sampath <ashwin.manning@gmail.com>